NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

RACE
Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2024
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED | MATHEMATICS | M.M. 62
REVISION ASSINGMENT # 04 | (P & C, BT) | TIME : 60 MIN.

SECTION–I

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. If the domain and range of $f(x) = {}^{9-x}C_{x-1}$ contains m and n elements respectively, then -
(A) m = n (B) m = n + 1 (C) m = n – 1 (D) m = n + 2

2. The number of ways in which all the letters of the word 'TANATAN' can be arranged so that no two alike letters are together
(A) 24 (B) 36 (C) 38 (D) 40

3. Number of ordered triplets (a, b, c) of positive integers less than 10, for which the product abc is divisible by 20, are -
(A) 48 (B) 102 (C) 54 (D) 51

4. The coefficient of x^4 in the expansion $(e^x - 1)^2$ is :
(A) $\frac{3}{8}$ (B) $\frac{5}{12}$ (C) $\frac{7}{12}$ (D) $\frac{5}{24}$

5. The sum of the series $(1^2+1)1! + (2^2+1)2! + (3^2+1)3! + ... + (n^2+1)n!$ is :
(A) $(n+1)((n+2)!)$ (B) $n(n+1)!$ (C) $(n+1)(n+1)!$ (D) none of these

6. The value of ${}^{3n}C_n - {}^nC_1 \cdot {}^{3n-3}C_n + {}^nC_2 \cdot {}^{3n-6}C_n - {}^nC_3 \cdot {}^{3n-9}C_n + ...$ is equal to :-
(A) 1 (B) 2^n (C) 3^n (D) $3^n - 2^n$

7. The coefficient of x^4 in the expansion of $(1+x)^{15} + (1+x)^{16} + (1+x)^{17} + ... + (1+x)^{50}$ is-
(A) ${}^{51}C_5 - {}^{15}C_5$ (B) ${}^{51}C_5$ (C) ${}^{51}C_5 - 1$ (D) ${}^{52}C_6 - {}^{15}C_4$

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

8. If $\sum_{r=0}^{n} \frac{r}{{}^nC_r} = \sum_{r=0}^{n} \frac{n^2 - 4n + 6}{2 \cdot {}^nC_r}$, then -
(A) n = 1 (B) n = 2 (C) n = 3 (D) n = 4

9. If set A contains m elements, B and C are two subsets of A such that n(B) = n(C) ; $B \cup C = A$; $n(B \cap C) = 1$
then identify the correct statements :-
(A) m must be odd
(B) m must be even
(C) Number of ways in which subset B and C are to be formed is $m \cdot {}^{m-1}C_{\frac{m-1}{2}}$
(D) Number of ways in which subsets B and C are to be formed is $\frac{1}{2} \cdot {}^{m-1}C_{\frac{m-1}{2}}$

10. If the expansion of $(2x - 3y + z)^{10} = \sum_{\substack{0 \le \alpha,\beta,\gamma \le 10 \\ \alpha+\beta+\gamma=10}} \frac{10!}{\alpha!\beta!\gamma!} (2x)^\alpha (-3y)^\beta (z)^\gamma$, then -
(A) the expansion will contain 66 terms
(B) $\sum_{\substack{0 \le \alpha,\beta,\gamma \le 10 \\ \alpha+\beta+\gamma=10}} \frac{10!}{\alpha!\beta!\gamma!} = 3^{10}$
(C) the coefficient of $x^2 y^3 z^5$ will be $\left(-\frac{9 \cdot 10!}{5!}\right)$
(D) $\sum_{\substack{0 \le \alpha,\beta,\gamma \le 10 \\ \alpha+\beta+\gamma=10}} \frac{10!}{\alpha!\beta!\gamma!} 2^\alpha (-3)^\beta = 0$

For More Material Join: @JEEAdvanced_2025

11. The number of selections of fruits from 4 mangoes, 3 oranges, 2 papayas and 2 pineapples will be [If fruits of same type are treated as same] -
(A) 179 (when atleast one fruit is chosen)
(B) 175 (atleast 2 fruits are chosen)
(C) 15 (at most 2 fruits are chosen with atleast one fruit)
(D) 15 (none or more fruits but not more than 2 fruits are chosen)

Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Question 12 to 14

If a man climbs either one step or k steps at a time then total number of ways in which he can climb up a staircase of 3k steps (starting from the bottom) where $k \geq 2$ is denoted as A(1, k)

12. A(1, 2) is equal to :-
(A) 10 (B) 11 (C) 12 (D) 13

13. If A(1, k) = 26 then k is equal to :-
(A) 2 (B) 3 (C) 4 (D) k

14. If $A(1, k+1) - A(1, k) = \lambda_k$ then $\lambda_{k+1} - \lambda_k$ is equal to :-
(A) 1 (B) 2 (C) 0 (D) 4

SECTION–II

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. The remainder when $\sum_{r=1}^{10}\left({}^{10}C_r\right)(r-5)^3$ is divided by 10, is

2. If $n \in N$, then the remainder when $(57)^{n+3} + (79)^{n+1} + (46)^n$ is divided by 11 is -

3. A gentleman gives a dinner party to 5 guests to be selected from 10 friends. The number of ways of forming a party of 5, given that two particular friends will not attend party together -

4. If the sum of all the odd divisors of 10!, which are of the form 3n + 1 is S_1 and which are of the form 3n – 1 is S_2. Then the remainder obtained by dividing S_1 by S_2 is

REVISION ASSINGMENT # 03 (COMPLEX NUMBER) MATHEMATICS

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	D	B	D	B	D	D	D	C	A,B,C,D	B,D
	Q.	11	12	13	14						
	A.	B,C	C	B	C						
SECTION-II	Q.	1	2	3	4						
	A.	4.50	1.00	5.00	100.00						

For More Material Join: @JEEAdvanced_2025